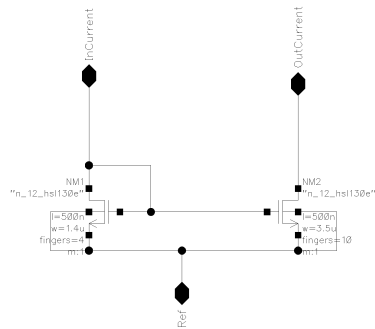
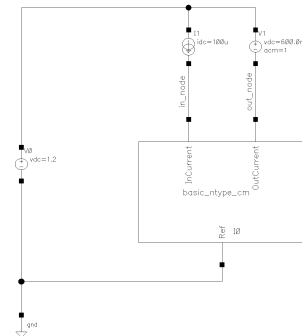


1 CAD Laboratory Session n°1 - Analysis and design of CMOS current mirrors

The first thing shown below are the schematic and the test bench used to analyze the current mirror and its behavior (see Fig. 1)



(a) Schematic of the CM



(b) Test bench used for the CM

Figure 1: Analysis of the CM

1.1 DC and AC analysis of the basic current mirror discussion

1.1.1 DC Analysis

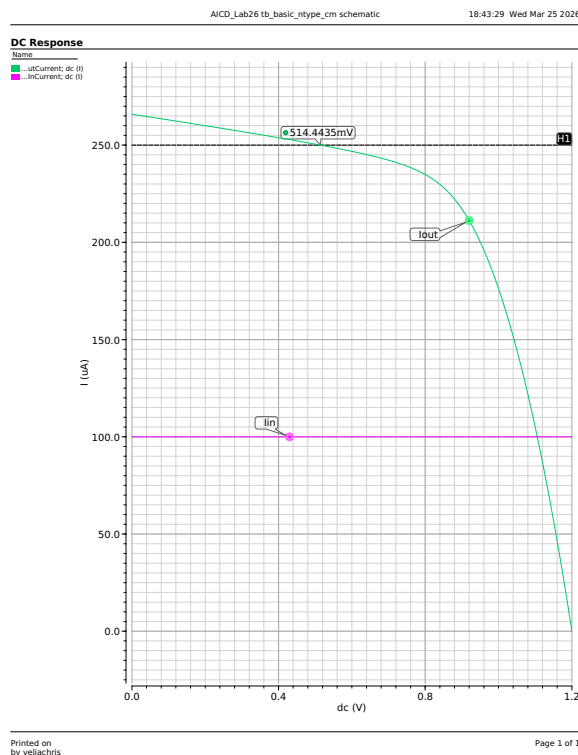


Figure 2: DC simulation

Looking at Fig.2, it can be seen the behavior of the output current while V_{IN} changes. Initially while V_{IN} is still small, the output current decreases almost linearly due to the channel length modulation, due to the components not being ideal. In this stage, the transistor M_2 is in saturation region. As V_{IN} increases, M_2 approaches the triode region where the output current drops faster and non linearly because the transistor acts more like a resistor, due to the fact that the transistor can no longer sustain a high V_{DS} . The reason why the identity holds only in this point ($V_{in} = 514 \text{ mV}$) is because the transistor M_2 is in saturation region and the current decreases linearly, as said above.

1.1.2 AC Analysis

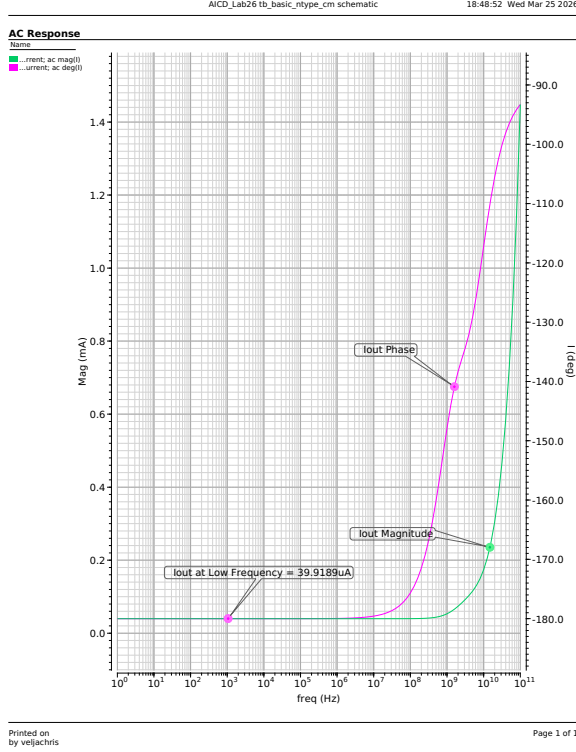


Figure 3: AC simulation

In the AC analysis, the output voltage is driven by an AC voltage signal of 1 V. Thanks to Fig.3, it is possible to evaluate the resistance r_o by extracting the value of I_{out}^{AC} at low frequency.

$$r_o = \frac{v_{ds}}{i_o} = \frac{1}{I_{out}^{AC}} = \frac{1}{39.9189\mu} = 25.05 \text{ k}\Omega$$

However, the expected value of the resistance is slightly different, so we calculate using the following formula :

$$r_o = \frac{\eta_n \cdot L_2}{I_{out,ac}}, \quad L_2 = 500nm, \quad \eta_n = 1.1 \cdot 10^7 \frac{V}{m} \Rightarrow r_{o2} = 22k\Omega$$

$$I_{out,ac} = 250\mu A$$

1.2 Simulation of the effect of process spread and device mismatch

To assess the circuit's robustness under extreme process conditions, it was conduct a simulation in ADE XL using three process corners:

- Nominal (T = 27C) : Standard operating point
- Slow-Slow (SS) at 80C : Both transistors exhibit slower performance
- Fast-Fast (FF) at -20C : Both transistors operate faster than usual

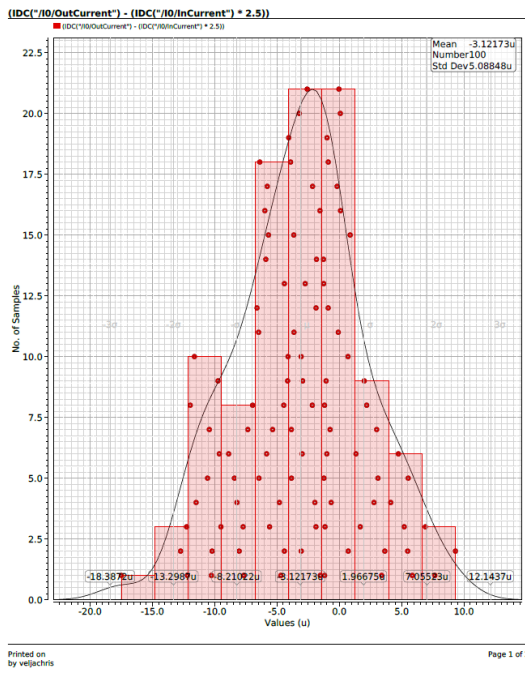
This analysis focuses on evaluating the CM input voltage and the CM output current error to ensure stable circuit performance under all conditions. The results are presented in the table below.

Output	Nominal TT	Slow-Slow SS	Fast-Fast FF
ΔI_O	-3.096μ	-5.516μ	-1.155μ
V_{DC}	$686m$	$751.4m$	$630.8m$

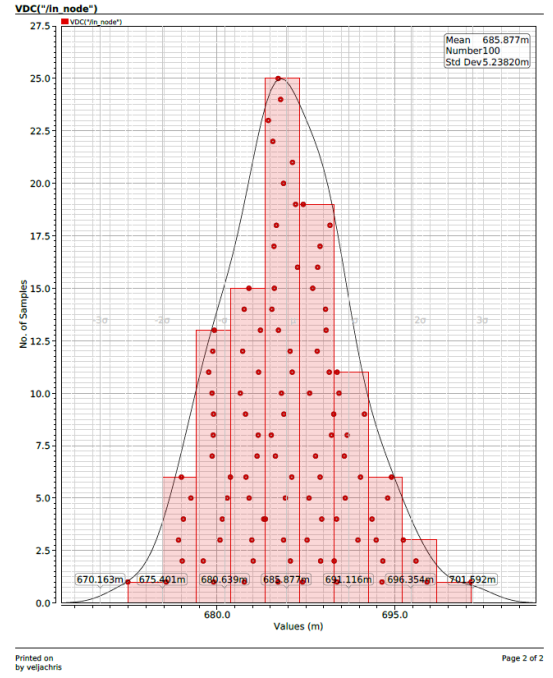
Table 1: Corner Simulations Results

It is shown that at lower temperature, the circuit works faster. This is due to the transconductance factor $k'_n = \mu_n C_{ox}$, which is proportional to the mobility carrier factor μ_n . At lower temperature, the silicon atoms are still, so electrons and holes are free to move around the crystal. At higher temperature, the silicon atoms begin to move more violently, causing electrons and holes to collide with the crystal lattice (phonon scattering), making the circuit slower, causing the current to decrease.

After the corner simulations, a Monte Carlo simulation is performed in order to understand how the statistical variations in the transistor parameters, due to the manufacturing process, affect the overall performance of the current mirror (see Fig. 38).



(a)



(b)

Figure 4: Monte Carlo Simulation

Standard deviation of the Output Current : $5.08848\mu A$. Std deviation for the voltage at the input node : $5.2382mV$.

For the standard deviation, we use the following formula :

$$\sigma\left(\frac{\Delta I_o}{NI_I}\right) = \sqrt{\sigma^2\left(\frac{\Delta k'_n}{k'_n}\right) + \left(\frac{g_{m2}}{I_{D2}}\right)\sigma^2(\Delta V_{TN})} = \sqrt{\frac{A_{\Delta k'_n}}{W_2 L} + \left(\frac{2}{V_{OV}}\right)^2 \frac{A_{\Delta V_{TN}}}{W_2 L}}$$

1.3 Wide-Swing Cascode Current Mirror

Lastly, it is asked to design a Wide-Swing Cascode Current Mirror. At Fig. 5 are shown the schematic and the test bench used to analyze it.

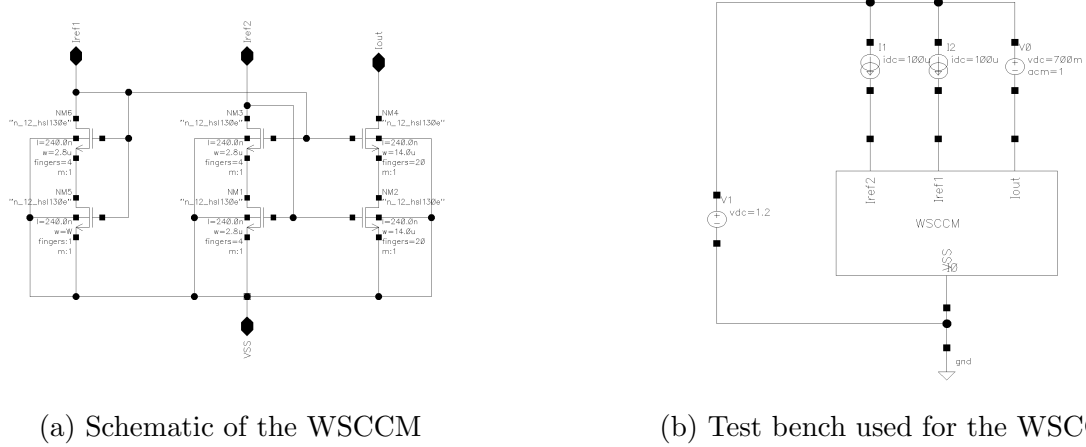


Figure 5: Analysis of the Wide Swing Cascode Current Mirror

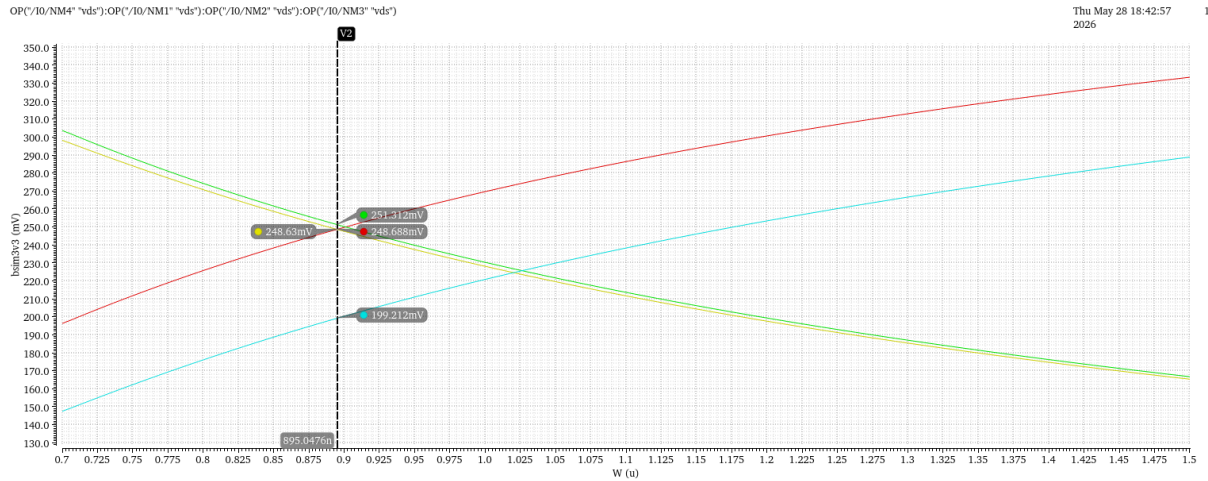


Figure 6: Drain-Source voltage of transistors M_1 , M_2 , M_3 and M_4

After a sweep done with the value of the gate width of transistor M_5 (see Fig. 6), it was possible to notice how it is impossible to achieve $V_{DS} = 0.25V$ for M_1 , M_2 , M_3 , M_4 since the drain of transistor three is connected to the gate of the first, so with the current value of W_1 it cannot be achieved.

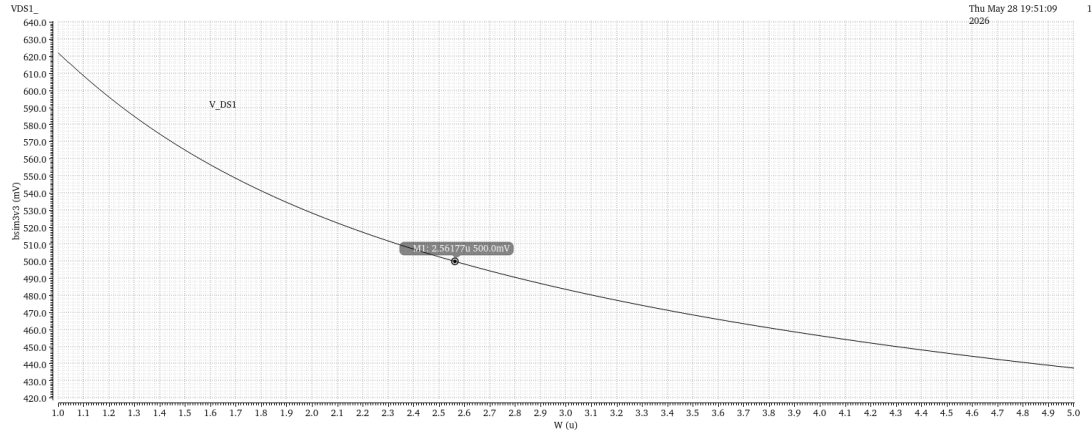


Figure 7: Sweep of transistor M_1 gate width

In Fig. 7, it can be seen when $V_{DS} = 0.25$ V. But, after doing again the simulation sweeping W_5 , the specification was again not met, because the drain-source voltage of M_3 was higher than 0.25 V. The gate width of transistors M_1, M_3 and M_6 were increased, and so M_2 and M_4 by a factor of 5:

$$W_1 = W_3 = W_5 = 2.8 \mu m$$

$$W_2 = W_4 = 5 \cdot W_1 = 14 \mu m$$

The DC sweep simulation for the right value of W_5 was made and, as can be seen in Fig. 8, having $W_5 = 770$ nm meets the specifications (see Fig. 9)

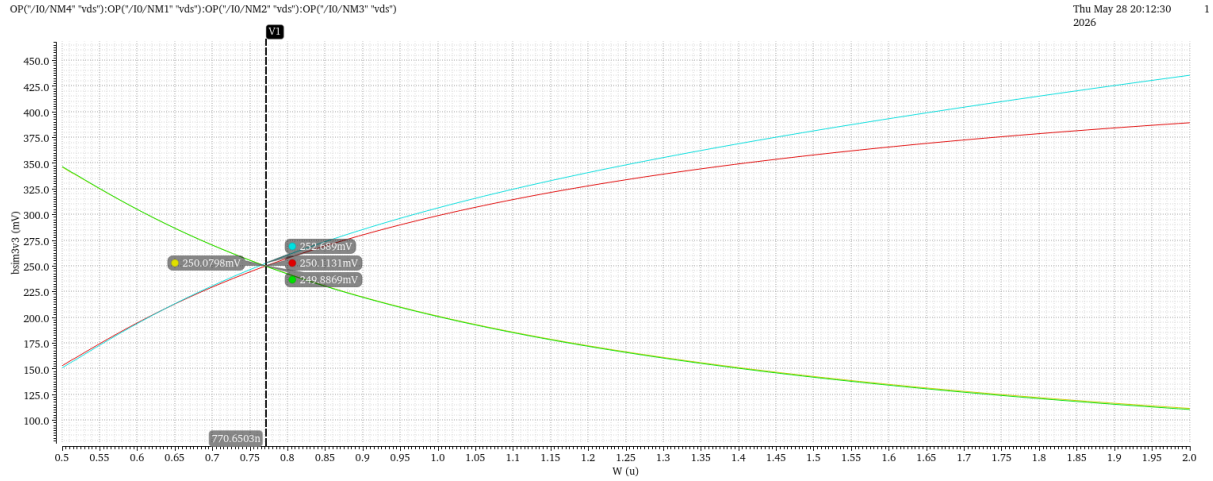


Figure 8: W_5 sweep

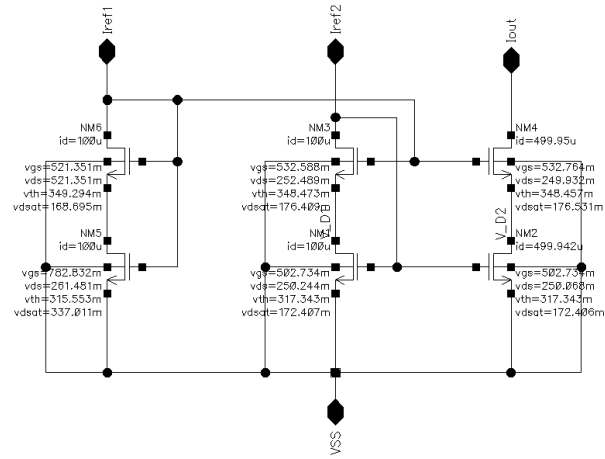


Figure 9: WSCCM DC Operating Points

After designing the WSCCM, the DC and AC simulations were again performed, as it is possible to see in Fig. 10 and Fig. 11

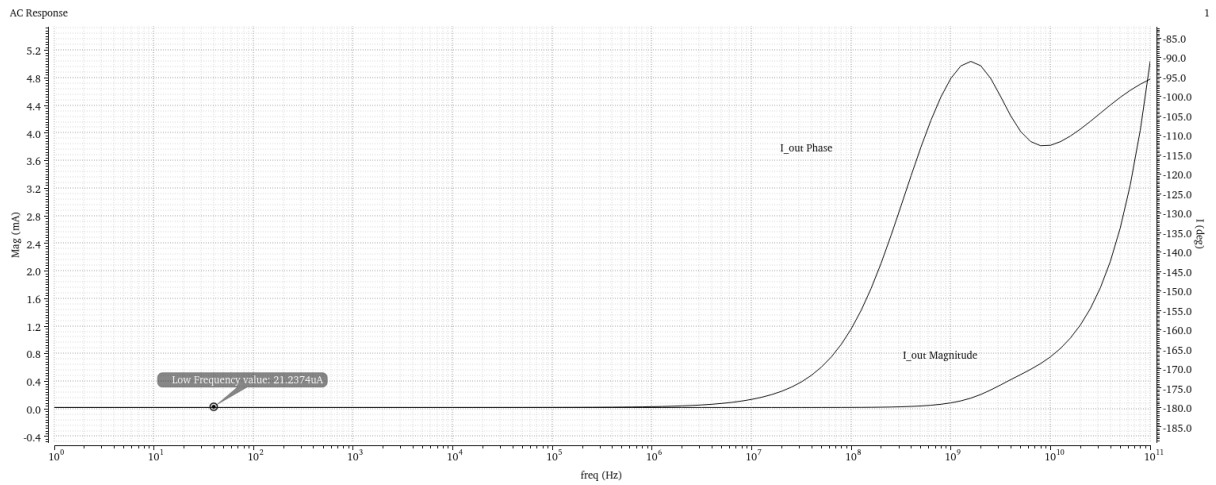


Figure 10: AC Simulation after the design

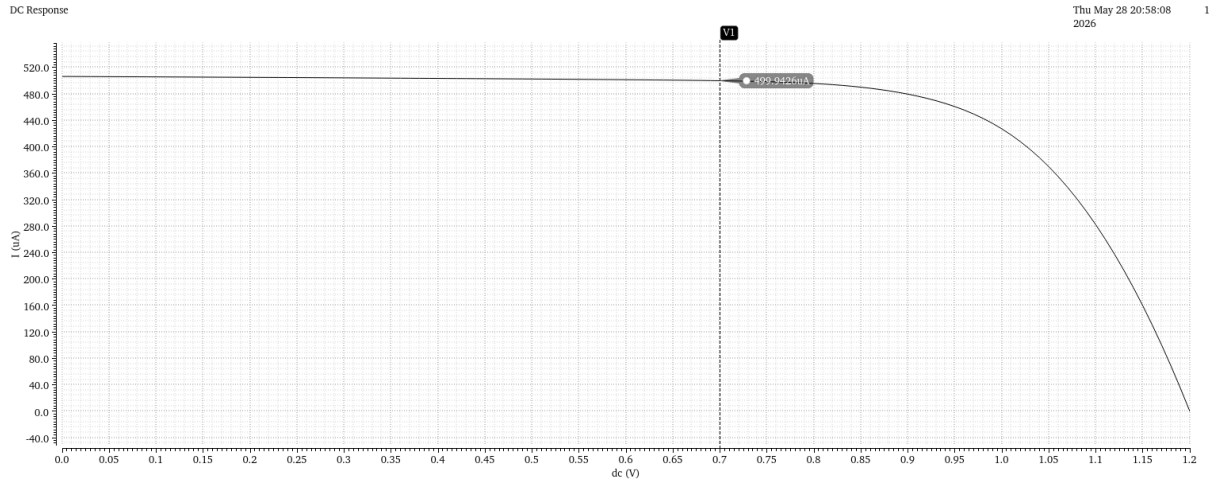


Figure 11: DC Simulation after the design

1.3.1 Output current error estimation

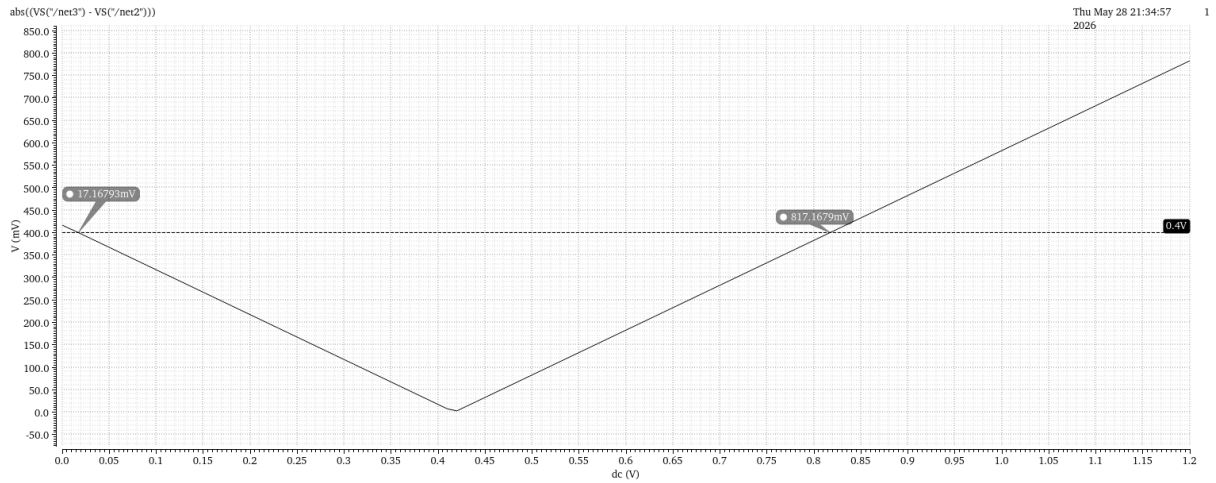


Figure 12: Plot of $|V_{OUT} - V_{IN}|$

The estimation for the output current error, was made when $|V_{OUT} - V_{IN}| = 0.4$ (see Fig. 12). Markers at those position were placed in the plot of the output current in order to see the error at those values (Fig. 13)

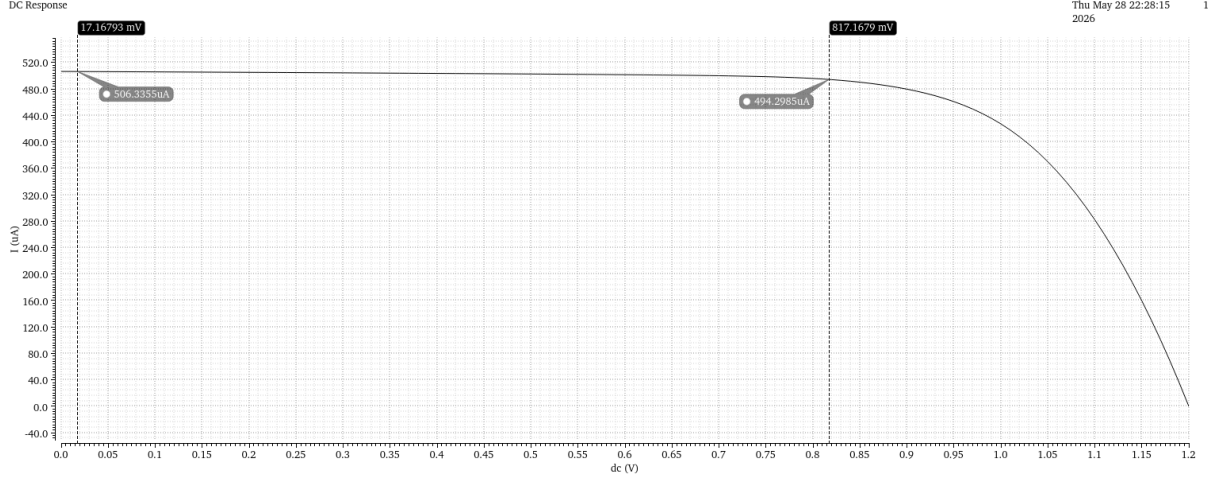


Figure 13: Estimation for the output current error

$$\text{Error at } 17.16794 \text{ mV} : | 506.3355 \mu A - 500 \mu A | = 6.3355 \mu A$$

$$\text{Error at } 817.1679 \text{ mV} : | 494.2985 \mu A - 500 \mu A | = 5.7015 \mu A$$

1.3.2 Adjustment of R_o

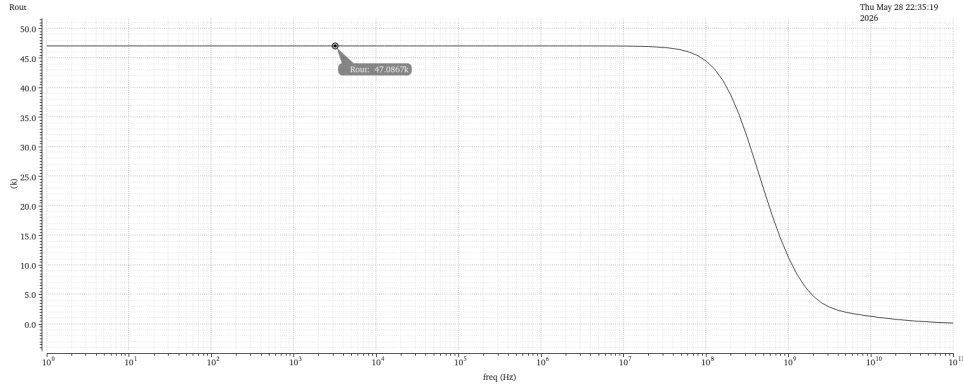


Figure 14: Estimation of R_o

To find out the value of R_o , an AC simulation was done (see Fig. 14) with the help of the output port voltage and output current, and found out

$$R_o = 47.0867 \text{ k}\Omega$$

Since the specification required that it be equal to $100 \text{ k}\Omega$, to do so, all transistors were increased by a factor of three, also having all gate lengths equal to 720 nm . The final value of the output resistance is (see Fig. 15)

$$R_o = 100.294 \text{ k}\Omega$$

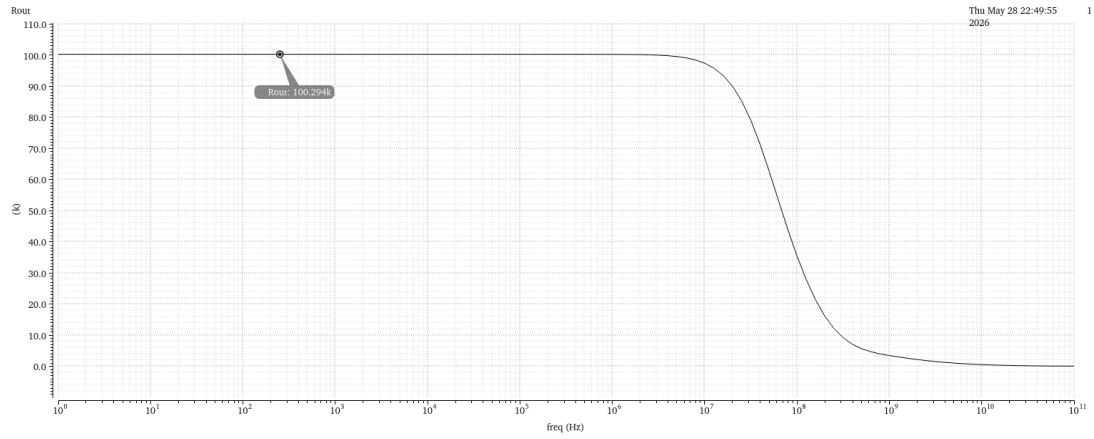


Figure 15: R_o within the specification